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Supplementary Material for: The Loyola Index Family and the Resolution Limit of Degree Imbalance

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This supplement accompanies the paper "The Loyola Index Family and the Resolution Limit of Degree Imbalance". Section, table, figure, equation and proposition numbers without the prefix S refer to the main paper, and data, scripts and the verifier are available at `https://github.com/Singati2/LOYOLA-PAPER`.

S1 Octane data provenance

Table 4 of the main paper reports the Pearson correlation r between each index and each of the five physicochemical properties of the 18 octane isomers. The properties are the boiling point T_B, the enthalpy of formation ΔH_f, the enthalpy of vaporisation ΔH_{vap}, the entropy S and the acentric factor ω. The property data were traced to the NIST WebBook [11],

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and for each value the provenance file records the WebBook entry or, where the WebBook lists no value, the compilation consulted. Acentric factors, which NIST does not list, are the exception. They are compiled values whose original source we could not identify. They agree with the values of the Korea Thermophysical Properties Data Bank (KDB) [9], consulted through Cheméo [4], and with the octane set reproduced in [5] to within 0.005, except for 2,2,3,3-tetramethylbutane (0.247 against 0.251 and 0.255). Agreement with a secondary compilation is not independent experimental confirmation when both sources derive from the same underlying measurement. Most ΔH_f values rest on one combustion study and most ΔH_{vap} values on one calorimetric study, as recorded with the number of independent determinations behind each value in the claim provenance ledger of the reproducibility package. The same five properties form the octane benchmark used in [14]. T_B is in °C, ΔH_f and ΔH_{vap} are in $\mathrm{kcal\,mol^{-1}}$ (the latter at 298 K), S is in $\mathrm{cal\,mol^{-1}K^{-1}}$, and ω is dimensionless. Values for ΔH_{vap} and entropy S vary slightly across compilations, and all comparisons here are made within this single, internally consistent dataset. Calculations use this author-compiled property table attributed to the NIST WebBook. All 90 molecule-level values were traced against authoritative sources, with the NIST WebBook consulted first. The values were classified under a strict taxonomy. In it, “verified” requires agreement at the value’s reported precision, and a converted value exactly half a display unit away counts as agreement at reported precision. Of the values, 3 match exactly, and 29 match a specific recorded determination at reported precision after unit conversion, with the gas phase confirmed for formation enthalpies and entropies. A further 39 agree within the audit tolerances, the source’s stated uncertainty or the span of several determinations without matching any one of them at reported precision. 10 lie within explained variation among authoritative determinations, namely 1947 API Project 44 entries superseded by later TRC revisions. 8 could not be verified against a reachable authoritative value, seven of these being gas-phase entropies absent from the NIST WebBook. For them the 1947 API Project 44 tables give values differing by 0.07–2.2 units, as they do for isomers whose later revisions were confirmed. The remaining one is

the 298 K vaporisation enthalpy of 2,2,3,3-tetramethylbutane. This compound is a solid at 298 K (triple point 374 K), and the NIST WebBook lists standard vaporisation enthalpies of 42.94 and 42.91 kJ mol^{-1} (about 10.26 kcal mol^{-1}) together with sublimation enthalpies. The benchmark value used here, 8.41 kcal mol^{-1} (35.19 kJ mol^{-1}), is that of the standard octane benchmark compilation and matches no listed determination. Its physical reference state is therefore unresolved. Section S3 reports the ΔH_{vap} analysis with this isomer excluded. One value varies between compilations by more than the audit tolerance, the acentric factor of 2,2,3,3-tetramethylbutane (0.247 against 0.251 in KDB and 0.255 in [5]), while the acentric factor of 2,2,4-trimethylpentane (0.305 vs. compiled 0.303) agrees only within tolerance. Substituting the alternative compiled acentric factors changes several of the ω correlations in Table 4 of the main paper at the third decimal (e.g. M_2 moves from -0.988 to -0.986). It does not change the property-wise winning descriptor or any substantive interpretation, and the full sensitivity computation and the machine-readable provenance trace accompany the reproducibility package. For 3,3-dimethylhexane, 3-ethyl-3-methylpentane, 2,2,3-trimethylpentane and 2,3,3-trimethylpentane, values of ΔH_{vap} that differ from the NIST WebBook by 0.05–0.13 kcal mol^{-1} circulate in octane benchmark tables. We use values consistent with NIST (Table S5 and Section S8). Two are exact conversions of NIST determinations (8.97, 9.08), and two lie within the span or stated uncertainty of the NIST determinations. These are 8.83, which also equals the API Project 44 value at the displayed precision, and 8.90, which equals the compiled octane dataset value 8.897 at the displayed precision. Seven gas-phase entropies could not be verified against a primary source (see above). Four entropies (n-octane, 2,2-dimethylhexane, 2,2,4- and 2,3,3-trimethylpentane) differ from those of a widely used octane QSPR compilation by 0.04–2.29 units, the 2, 3, 3-trimethylpentane value 102.06 being a compiled-dataset value (see Section S8). Substituting the compilation values changes all twelve S correlations in Table 4 of the main paper, by up to 0.043 in $|r|$, but it does not change the winning descriptor (HM). Apart from the four corrected vaporisation enthalpies and the corrected transcription of the 2,3,3-trimethylpentane entropy (Section S8,

item (ii)), the benchmark values are preserved as supplied for comparability.

Correlation details for Table 4 of the main paper. The best correlation per property is split among the reductions and the pure-γ points, with the modified second Zagreb index $LO(-1,0,0)$ strongest on T_B and ΔH_f. The hyper-Zagreb index $LO(0,2,0)$ is strongest on S, the second Zagreb index $LO(1,0,0)$ on ω, and the fixed pure-γ benchmark point $LO(0,0,1)$ on ΔH_{vap} ($|r| = 0.984$, with $LO(0,0,2)$ second at 0.981). At the exact-reduction triples the Loyola correlations coincide with the corresponding classical indices to floating-point precision, as they must. For each property, a descriptive post-selection paired bootstrap interval for the observed top-two $|r|$ difference included zero ($B = 10\,000$, resampling the 18 isomers jointly, `default_rng(1)`). These intervals were not multiplicity-adjusted across the five properties and do not establish equality or equivalence. Individual bootstrap half-widths range from 0.016 to 0.096 across the headline cells (percentile bootstrap, $B = 10\,000$, `default_rng(0)`). The value of $LO(0,0,1)$ on ΔH_{vap} is that of a fixed benchmark point and involves no tuning. Recomputing each bolded correlation with single isomers deleted shifts $|r|$ by at most 0.027, which indicates that no single isomer drives the leading correlations.

Descriptor redundancy. The descriptors are strongly collinear. This limits what any small $|r|$ difference in Table 4 of the main paper can mean. Across the twelve compared descriptors on the 18 isomers, fourteen descriptor pairs have $|r| \geq 0.99$, the extreme being R versus $H/2$ at $r = 0.9997$. A principal component analysis of the standardised 18×12 descriptor matrix places 90.3% of the variance on the first component and 99.05% on the first two. The effective rank (participation ratio of the eigenvalue spectrum) is 1.2, so the twelve-index comparison set is essentially a two-dimensional descriptor space. Differences in the third decimal of $|r|$ between near-collinear descriptors should not be read as meaningful rankings. Figure S1 shows the signed-r heatmap of the compared indices against the five properties, displaying the values of Table 4 of the main

paper.

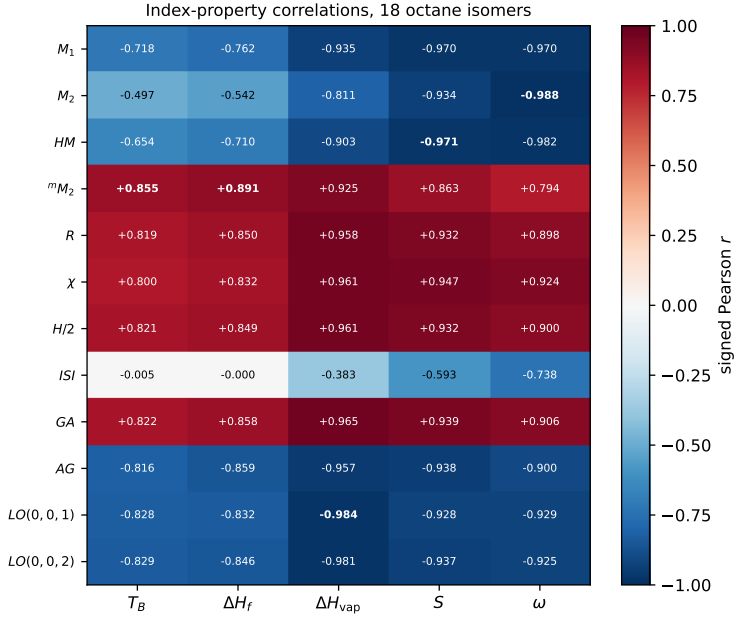


Figure S1. Signed Pearson r between the Loyola reductions, the fixed pure-γ benchmark points and the five octane physicochemical properties on the 18 isomers. The entry -0.000 (ISI, ΔH_f) is -0.0002.

S2 Local parameter tuning

The benchmark of Section 6.1 of the main paper tunes no parameters and compares only the exact reductions and the two fixed pure-γ benchmark points. It is nonetheless natural to ask whether small, data-driven adjustments of (α, β, γ) near the best-correlating indices could improve the fit, and this section examines that question as an exploratory check. We take the four anchors carrying the strongest column correlations in Table 4 of the main paper, namely $M_2 = LO(1,0,0)$, $HM = LO(0,2,0)$, ${}^mM_2 = LO(-1,0,0)$, and the pure-γ point $LO(0,0,1)$. For each anchor we draw 50 random parameter triples within ± 0.3 (to two decimal places)

of the anchor and correlate each against the five octane properties. The four pools are drawn, in the order listed, from one pseudo-random stream (NumPy `default_rng(42)`), each coordinate offset being a uniform draw on $[-0.3, 0.3]$ rounded to two decimals. The percentile bootstrap uses `default_rng(0)`. For every anchor–property cell, Table S1 reports the anchor correlation $|r|_0$, the best of the 50 perturbations in sample $|r|_{50}$, the in-sample gain Δ_{in} and the percentile-bootstrap half-width ($B = 10\,000$). It also reports the leave-one-out correlation of the anchor LOO_0 and a *nested* leave-one-out correlation LOO_{50}. In the nested version the best of the 50 candidates is selected, within each of the four globally selected anchor pools, using the 17 training isomers of each fold only. It is then evaluated on the held-out one. The resulting out-of-sample gain is Δ_{out}. The anchors themselves were chosen from the complete-data correlations of Table 4 of the main paper, and the 18 pooled fold predictions share heavily overlapping training sets. These values are therefore descriptive rather than independent replications.

Table S1. Decimal parameter tuning near the four best-correlating anchors on the 18 octane isomers (50 random triples within ±0.3 of each anchor). The columns give the anchor correlation $|r|_0$, the best-of-50 in sample $|r|_{50}$, the in-sample gain Δ_{in} and the percentile-bootstrap half-width ±boot ($B = 10\,000$). They then give the leave-one-out correlation of the anchor LOO_0, the foldwise nested leave-one-out correlation of the tuned family LOO_{50}, computed within the globally selected anchor pool, and the out-of-sample gain $\Delta_{\text{out}} = \text{LOO}_{50} - \text{LOO}_0$. Gains are computed from full-precision values and rounded once, so a displayed gain may differ by one unit in the last place from the difference of the displayed operands.

Property	$\lvert r\rvert_0$	$\lvert r\rvert_{50}$	Δ_{in}	±boot	LOO_0	LOO_{50}	Δ_{out}
near $M_2 = LO(1,0,0)$							
T_B	0.497	0.598	+0.100	0.353	0.331	0.458	+0.127
ΔH_f	0.542	0.635	+0.093	0.358	0.395	0.540	+0.145
ΔH_{vap}	0.811	0.876	+0.066	0.185	0.768	0.838	+0.071
S	0.934	0.955	+0.021	0.073	0.909	0.935	+0.025
ω	0.988	0.995	+0.007	0.018	0.985	0.985	0.000
near $HM = LO(0,2,0)$							
T_B	0.654	0.740	+0.085	0.257	0.545	0.647	+0.102
ΔH_f	0.710	0.786	+0.076	0.255	0.645	0.741	+0.097
ΔH_{vap}	0.903	0.944	+0.041	0.082	0.871	0.921	+0.049
S	0.971	0.973	+0.002	0.030	0.961	0.957	−0.004
ω	0.982	0.989	+0.007	0.025	0.979	0.982	+0.003
near ${}^mM_2 = LO(-1,0,0)$							
T_B	0.855	0.864	+0.009	0.096	0.781	0.759	−0.021
ΔH_f	0.891	0.897	+0.006	0.087	0.865	0.837	−0.028
ΔH_{vap}	0.925	0.966	+0.041	0.068	0.900	0.953	+0.053
S	0.863	0.917	+0.054	0.192	0.810	0.878	+0.068
ω	0.794	0.881	+0.088	0.246	0.730	0.853	+0.123
near $LO(0,0,1)$							
T_B	0.828	0.835	+0.007	0.161	0.771	0.764	−0.007
ΔH_f	0.832	0.842	+0.010	0.152	0.800	0.807	+0.006
ΔH_{vap}	0.984	0.985	+0.001	0.016	0.978	0.978	−0.001
S	0.928	0.945	+0.017	0.046	0.887	0.915	+0.028
ω	0.929	0.945	+0.016	0.074	0.907	0.926	+0.019

Three observations can be made. First, every in-sample gain Δ_{in} is smaller than the corresponding single-descriptor bootstrap uncertainty scale, although this comparison is descriptive rather than a formal paired or post-selection test of the gain. Second, under the foldwise nested evaluation (within the globally selected anchor pools) these descriptive point-estimate gains do not vanish. Tuning near M_2 raises the out-of-sample correlation for T_B and ΔH_f by +0.13 and +0.15, and tuning near ${}^mM_2$ raises ω by +0.12. In every such case, however, the anchor was a *weak* descriptor for that property, and the tuned correlation only climbs part

way toward the level a strong reduction already attains. Tuning near M_2 lifts T_B to 0.46, whereas ${}^{m}M_2$ already reaches 0.78. Where the anchor is already the column winner (M_2 on ω, HM on S, $LO(0,0,1)$ on ΔH_{vap}), the nested gain for this pool seed is -0.0003, -0.004 and -0.001. These near-zero values are specific to the seed. Re-drawing the pools with seeds 0–99, the nested gain for M_2 on ω is small but positive in most seeds (median $+0.003$, positive in 97 of 100 seeds). The sweep is in the repository script `tuning_pool_seed_sweep.py`. Tuning a column winner therefore changes its nested correlation by at most a few thousandths. Third, tuning near the pure-γ points $LO(0,0,1)$ and $LO(0,0,2)$ does not exceed the best fixed index under nested leave-one-out on any of the five properties (Table S2). On ΔH_{vap} the best fixed index is the pure-γ point $LO(0,0,1)$ itself. There the tuned value exceeds it by 0.001 in sample (0.985 vs 0.984) and falls 0.001 below it under leave-one-out (both 0.978 at three decimals, the differences being those of full-precision values). These post-selection differences are negligible. On ΔH_f, S and ω the best fixed reduction is higher both in and out of sample. On T_B it is higher in sample, but under nested leave-one-out the tuned value is higher (0.764 against 0.745).

Table S2. Decimal tuning near the pure-γ points $LO(0,0,1)$ and $LO(0,0,2)$ versus the best fixed index in the tested comparison set for each octane property. The best fixed index is ranked by in-sample $|r|$. The decimal columns give the best over two pools of 50 random triples within ± 0.3 of $LO(0,0,1)$ and of $LO(0,0,2)$, in sample and under fold-wise nested leave-one-out within the globally selected pools. Here the $LO(0,0,1)$ pool is exactly the pool of Table S1, and the $LO(0,0,2)$ pool is the next 50 triples of the same `default_rng(42)` stream. Indices are ranked by in-sample $|r|$ rather than LOO, as the latter is inflated by degenerate fits for near-zero-correlation indices. The $\mathrm{LOO}_{\mathrm{fix}}$ column is not nested. With the fixed choice nested in the folds it becomes 0.745 for T_B and 0.950 for S, and tuning then exceeds the fixed index only on T_B (0.764). In sample it exceeds it only on ΔH_{vap}, by 0.001, a negligible post-selection difference that does not establish superiority.

Property	best fixed	$\lvert r\rvert_{\mathrm{fix}}$	$\mathrm{LOO}_{\mathrm{fix}}$	$\lvert r\rvert_{\mathrm{dec}}$	$\mathrm{LOO}_{\mathrm{dec}}$
T_B	${}^{m}M_2$	0.855	0.781	0.835	0.764
ΔH_f	${}^{m}M_2$	0.891	0.865	0.850	0.815
ΔH_{vap}	$LO(0,0,1)$	0.984	0.978	0.985	0.978
S	HM	0.971	0.961	0.945	0.915
ω	M_2	0.988	0.985	0.945	0.926

Local tuning raised several nested leave-one-out point estimates for weak anchors. Except for the 0.001 in-sample ΔH_{vap} margin and octane T_B under nested leave-one-out (0.764 against 0.745), it did not exceed the best fixed benchmark in the tested comparison set. No formal superiority claim is made. All tabled nested values correspond to positive signed prediction–observation correlations with positive predictive Q^2. A constant-descriptor control confirms that absolute pooled correlation alone can be inflated for degenerate descriptors (cf. the caveat in the caption of Table S2).

S3 Matched-budget comparison

The analyses above tune around fixed anchors and are descriptive. The question raised by the Gutman–Milovanović framing is whether, given identical search budgets, freeing γ improves out-of-sample prediction beyond an optimised parent $M_{\alpha,\beta}$. This is tested with a fully nested

Table S3. Matched-budget nested leave-one-out comparison of the optimised parent $M_{\alpha,\beta}$ (GM) and the optimised Loyola family (γ free) on the five octane properties. Panel (a) is a *single illustrative candidate stream* (seed 12345, budget 200), where Q^2 is the outer predictive coefficient and RMSE the outer root-mean-square error. "LO better" counts outer folds (of 18) in which the Loyola model had the strictly smaller absolute error. Panel (b) summarises the 100-seed sweep (seeds 0–99), giving the median ΔQ^2 with its [minimum, maximum], and the number of seeds in which $\Delta Q^2 > 0$, at budgets $B = 200$ and $B = 500$. All selection is repeated inside every outer fold, and all values are full-precision results rounded once.

(a) single candidate stream: seed 12345*, budget* 200

Property	Q^2_{GM}	Q^2_{LO}	ΔQ^2	$\mathrm{RMSE}_{\mathrm{GM}}$	RMSE_{LO}	LO better
T_B	+0.613	+0.748	+0.136	3.700	2.983	10/18
ΔH_f	+0.855	+0.549	−0.306	0.463	0.816	4/18
ΔH_{vap}	+0.884	+0.971	+0.086	0.127	0.064	13/18
S	+0.906	+0.878	−0.029	1.363	1.558	7/18
ω	+0.979	+0.956	−0.022	0.005	0.008	5/18

(b) 100 *candidate streams: seeds* 0–99

	median ΔQ^2 [min, max]		seeds with $\Delta Q^2 > 0$	
Property	$B = 200$	$B = 500$	$B = 200$	$B = 500$
T_B	+0.010 [−0.524, +0.373]	+0.066 [−0.307, +0.231]	53/100	64/100
ΔH_f	−0.120 [−0.408, +0.206]	−0.095 [−0.373, +0.118]	14/100	24/100
ΔH_{vap}	+0.069 [+0.034, +0.102]	+0.068 [+0.024, +0.102]	100/100	100/100
S	−0.012 [−0.140, +0.038]	−0.010 [−0.098, +0.031]	34/100	31/100
ω	−0.010 [−0.074, +0.016]	−0.006 [−0.052, +0.012]	26/100	28/100

leave-one-out design. For each property and each outer fold (one isomer held out), 200 candidate parameter vectors are drawn once from Uniform$(-2, 2)$ per coordinate. The candidates are triples (α, β, γ) for the Loyola model and pairs (α, β) with $\gamma = 0$ for the parent. The budget is identical. Candidates are rounded to three decimals and drawn from NumPy `default_rng(12345)` for Table S3, Loyola triples first. On the 17 training isomers only, every candidate is scored by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression. The best candidate is selected, the regression is refit on all 17, and the held-out isomer is predicted. No quantity computed on the held-out isomer enters selection or fitting at any point.

Once the parent is searched over a wide box, freeing γ gives no consistent gain (Table 5 and Section 6.2 of the main paper). The comparison in the box $(-2, 2)$ is reported first because it shows how the apparent gain arose, and in that box the result is mixed (Table S3). For the illustrative seed, freeing γ improves nested predictive performance on two of the five properties (T_B, $\Delta Q^2 = +0.136$, and ΔH_{vap}, $\Delta Q^2 = +0.086$) and degrades it on three (ΔH_f, S, ω). The ΔH_f degradation is driven by selection instability. The parent model selected the same (α, β) in all 18 folds there, while the three-parameter search selected widely varying triples (fold-to-fold standard deviation above 1.2 in both α and β). Selected γ values are interior to the search box (maximum $|\gamma| = 0.48$). The parent's selections are not. For ΔH_{vap} the parent chose $(\alpha, \beta) = (1.021, -1.993)$, on the boundary of the box, in 16 of the 18 folds, a choice whose consequences are examined in Section 6.2 of the main paper. On the one property where γ helps most stably, ΔH_{vap}, the identical triple $(-0.270, +0.271, +0.239)$ was selected in every fold. With $n = 18$ molecules these Q^2 differences carry substantial selection noise. We attach no inferential claim to them.

Because the comparison depends on a random candidate stream, the full nested design was repeated for one hundred seeds (0–99) at both budgets (200 and 500 candidates per model). This makes 1000 configurations in all. A paired control was run alongside, comparing the Loyola candidates against the *same* (α, β) coordinates with γ set to zero, which isolates the effect of γ from the independent candidate streams. Freeing γ improved nested predictive performance for ΔH_{vap} in every seed of this sweep (100/100 at both budgets, and 100/100 in the paired control). The median was $\Delta Q^2 = +0.069$ and $+0.068$, with range $+0.024$ to $+0.102$. It *usually* degraded performance for ΔH_f (86/100 and 76/100 seeds), while for T_B the split is close to even at budget 200 (53/100) but leans towards improvement at budget 500 (64/100). S and ω usually degrade slightly. These counts describe optimiser variance over random candidate streams on one fixed set of 18 molecules, which are never resampled. The counts therefore carry no inferential weight. In particular, 100/100 does not establish that γ improves ΔH_{vap} prediction for other molecules or data sets. The result is also sensitive to a single data cell, the ΔH_{vap}

value of 2,2,3,3-tetramethylbutane, whose physical reference state is unresolved (Section S1). The complete nested comparison was repeated with that isomer excluded (17 molecules, same seeds, budgets and candidate streams). Freeing γ then improved ΔH_{vap} in 93/100 seeds at budget 200 and 96/100 at budget 500 (paired control 95/100 and 96/100). The median was $\Delta Q^2 = +0.046$ and $+0.051$, with range -0.048 to $+0.098$. The gain survives the exclusion but is no longer sign-invariant. These comparisons share the box $(-2, 2)$, which truncates the parent along the direction $\beta \approx -2\alpha$. Section 6.2 and Table S6 show that the vaporisation-enthalpy gain shrinks to about zero once the parent is searched over a box that does not truncate it.

S4 Nonane replication

Two limitations of the octane analysis are its size (18 molecules) and its restriction to index models. Both were addressed under a protocol recorded in a git commit of the reproducibility package before any nonane value was collected. The identical nested design was applied to the 35 constitutional nonane isomers. Boiling points (35 values) and standard vaporisation enthalpies (34 values, none being listed for 3-ethyl-4-methylhexane) were taken from the NIST WebBook [11], with every value traced to its WebBook source page (retrieved on 3 October 2026). The two properties differ in evidential status. Twenty of the 35 boiling points are NIST averages of experimental determinations (13 quoted only to 1 K). Five are single NIST determinations and ten are the most recent primary determination listed by NIST (Deviation D1 of the protocol). All are experimental measurements, so the T_B test is experimentally grounded. For 3-ethyl-4-methylhexane NIST also lists 407.15 K (Lukes and Langthaler, 1957) under a separate entry for the dextrorotatory enantiomer, with an uncertainty of 2 K assigned by TRC against 0.5 K for the registered value. Enantiomers share one normal boiling point, so the two values are discrepant measurements. The second value lies 6.4 K below the registered 413.55 K, which is why the molecule is excluded from the pooled boiling-point cohort of Section 6.4 of the main paper. With it excluded, LO is better than GM

in 55/100 and 60/100 seeds and ridge regression gives $Q^2 = 0.679$. With 134.0 °C the figures are 61/100, 66/100 and 0.647. The verdict stays mixed. Of the vaporisation enthalpies, only 4 of 34 are calorimetric determinations and one is a NIST average. The other 29 come from two compilations. The first is the 1961 tabulation of Labauf, Greenshields and Rossini [10]. It covers all 35 nonanes and 75 decanes in one short paper and, as far as we could determine, does not report individual calorimetric measurements. The second is a 1971 vapour-pressure handbook [21], under which NIST lists 25 values as Reid (1972), a review of that handbook [18]. Such tabulations may rest partly on structural-increment estimates, so agreement with them measures consistency with the compiled values rather than independent experimental validation. A degree-pair-based predictor may also be favoured if the compiled values were themselves derived from structural increments. The pre-registered decision rule called the γ effect *replicated* if the Loyola model beat the parent in at least 80 of 100 seeds at both budgets with positive median ΔQ^2. Four baselines were added, each fitted entirely within the outer training fold. They are the training-fold mean, the best of the ten classical indices of Table 4 of the main paper, and the irregularity index $IRLA$. The fourth is ridge regression on the full edge-degree-pair count vector, with its penalty chosen by inner leave-one-out, and the resulting Q^2 values are in Table S7.

On the nonane replication set the pre-registered verdicts are as follows. For ΔH_{vap} (compilation values) the γ effect met the pre-registered optimiser-stability criterion and *replicates*, LO being better than GM in 95/100 and 98/100 seeds, with median $\Delta Q^2 = +0.048$ and $+0.053$. For T_B the verdict is *mixed* (58/100 and 63/100). It is unchanged in direction under the pre-registered sensitivity analysis D4 of the protocol. That analysis replaces the NIST average for 2,3,3,4-tetramethylpentane, 413 ± 6 K, by the median of all eight listed determinations including a 1989 handbook value and a 1939 outlier, 414.62 K. LO is then better in 60/100 and 66/100 seeds, with median $\Delta Q^2 = +0.041$ and $+0.052$. The pre-registered protocol fixed the box of half-width 2, which, as Table S6 shows, binds the parent. In a box that does not, the nonane gain shrinks to about zero (median $\Delta Q^2 = -0.002$ under the hybrid control), and the pre-registered

verdict is reported as obtained, with this deviation recorded. The pattern behind the vaporisation result is the same on both sets. Models whose edge weight depends on imbalance only through q^2 inside the box of half-width 2 (the classical indices, including $GA = \sum_{uv} \sqrt{1 - q_{uv}^2}$, and GM) reach $Q^2 = 0.853$–0.888. Models with a term linear in q or with unrestricted counts ($IRLA$, ridge, $LO(0, 0, 1)$, LO) reach $Q^2 = 0.897$–0.957. This split holds only inside the narrow box. In the box of half-width 12 the parent alone reaches 0.942 and 0.892 (Table S6). The fixed point $LO(0, 0, 1)$ involves no tuning on the nonanes but was singled out on the octanes, and it matches or beats the tuned Loyola model on both sets (0.957 and 0.903 against medians 0.953 and 0.900). It is also the index selected in every fold for ΔH_{vap} when admitted to the fixed-index comparison. It is closely related to $IRLA$. One has $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2}\, IRLA(G) + O(\gamma^2)$ as $\gamma \to 0$, and on both sets the correlation between $LO(0, 0, 1)$ and $IRLA$ is 0.996. The baselines, however, also show the limit of the claim. Ridge regression on the full degree-pair count vector has the highest point estimate among the models on the nonanes, for both properties ($Q^2 = 0.927$ for ΔH_{vap}, 0.682 for T_B). The conditional intervals for LO minus ridge include zero for both (for ΔH_{vap} see the end of this section; the wider T_B interval is in `external/uncertainty_exploratory.csv`). On the octanes it matches the Loyola model on ΔH_{vap} and its point estimates are below the best index elsewhere. This is consistent with Proposition 5 of the main paper and the observation preceding it that every vertex-degree-based index is a function of the edge-degree-pair counts. The imbalance coordinate re-weights degree-pair information rather than adding to it, and one interpretable descriptor recovers most of what a model on all ten degree-pair counts achieves for the vaporisation enthalpy. It does not outperform such a model in point estimate. The specific exponential form is not essential either, since the parameter-free aggregate $IRLA$ comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). For this property a term linear in the imbalance suffices in point estimate inside the narrow box, but the wide-box parent shows that it is not necessary.

The nested design refits every model on the nonanes, so it tests whether the *procedure* replicates rather than whether an octane-trained model

transfers. An exploratory analysis followed. It was not pre-registered, and it was run in the box of half-width 2, so that it inherits the truncation of the parent. In it, the parameters selected on all 18 octanes were frozen and applied to the nonanes. With the parameters frozen and only the linear calibration refitted, the Loyola model beat the parent for ΔH_{vap} in 100/100 seeds at both budgets (median $Q^2 = 0.910$ and 0.916 against 0.864 and 0.865). It beat the parent for T_B in 65/100 and 82/100. With the octane calibration frozen as well, both models extrapolate poorly from C_8 to C_9. The Loyola model is then worse than the parent (median Q^2 between -2.22 and -0.72 for LO against 0.06 to 0.44 for GM). Because both comparisons inherit the truncated box, they do not bear on whether γ adds predictive information.

The seed counts above measure optimiser variance only. To indicate molecule-level uncertainty we also computed, post hoc, a paired bootstrap over molecules (2000 resamples) of the median-over-seeds ΔQ^2, conditional on the fitted outer predictions. Selection is not repeated within resamples. The intervals therefore do not account for uncertainty from refitting and model selection. The 95% intervals for LO minus GM on ΔH_{vap} are $[+0.014, +0.111]$ on the octanes but $[-0.006, +0.113]$ on the nonanes at budget 200 ($[-0.005, +0.119]$ at 500). Although the nonane gain met the pre-registered criterion, its interval narrowly includes zero, as does the interval for LO minus ridge on the nonanes, $[-0.101, +0.032]$. For T_B the intervals are wide ($[-0.212, +0.222]$ for LO minus GM on the nonanes). The one octane interval that excludes zero in the other direction is that for ΔH_f, where freeing γ is worse ($[-0.281, -0.044]$). All these intervals refer to the box of half-width 2. Given the search-box sensitivity of Table S6, no γ effect is established. The intervals are reported for completeness.

S5 Degeneracy across tree orders

Degeneracy [8] measures the failure of an index to discriminate non-isomorphic graphs, so lower values are better. On the 106 non-isomorphic

trees of order 10 we use the distinct-value convention

$$\text{Degeneracy}(\%) = 100\Big(1 - \tfrac{k}{N}\Big), \quad k = \#\{\text{distinct index values}\},\ N = 106,$$

i.e. the number $N - k$ of structures in excess of the distinct values, as a percentage of N. This is not the share of structures lacking a unique value, which is larger. For $LO(0, 0, 1)$ on this set, 45 of the 106 trees (42.5%) share their value with another tree. The pure-γ points $LO(0, 0, 1)$ and $LO(0, 0, 2)$ attain degeneracy 25.5% (79 distinct values), among the lowest of all compared indices and tied with R, GA and AG. The 106 trees realise only 79 distinct edge-degree-pair count vectors, so 25.5% is the minimum degeneracy attainable by *any* degree-pair-based index on this set, and the pure-γ points attain this floor. Theorem 2 of the main paper gives the reason and the order at which this stops. The first Zagreb index $LO(0, 1, 0)$ (83.0%) and the second Zagreb index $LO(1, 0, 0)$ (67.0%) are the most degenerate (Figure 1 of the main paper).

To test whether this floor attainment is peculiar to order 10, the same computation was repeated for all non-isomorphic trees of orders 7 through 12. It was also repeated, separately, for the molecular subsets with $\Delta \leq 4$ (Table S4), and the pure-γ points attain the degree-pair-profile floor at *every* order in both sets. The Randić index also attains the floor at every order in this range, while χ falls off the floor from order 10. Floor attainment therefore distinguishes the pure-γ points from the most degenerate classical indices (M_1, M_2) but not from R. Theorem 2 of the main paper turns this into an exact statement. The pure-γ points separate precisely the q-histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for $\Delta \leq 4$) and not beyond. Among the trees of order 13 the profiles $\{(1,2)^1, (1,3)^2, (1,6)^5, (2,3)^2, (2,6)^1, (3,3)^1\}$ and $\{(1,3)^3, (1,6)^5, (2,2)^1, (2,3)^2, (3,6)^1\}$ (exponents are edge multiplicities) share the q-histogram $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$, and there the pure-$\gamma$ points realise 566 distinct values against 570 profiles. Among molecular trees ($\Delta \leq 4$) the first failure occurs at order 16 (809 values against 810 profiles). The Randić index still attains the floor at order 13 (570 values) and falls below it at order 14 (1095 values against 1100 profiles, where the

pure-γ points give 1090).

Table S4. Discrimination across tree orders $n = 7$–12. The columns give the number of trees N, the number of distinct edge-degree-pair profiles (the floor attainable by any vertex-degree-based index), and distinct-value counts for representative indices. "mol." denotes the molecular subset $\Delta \leq 4$.

n	set	N	floor	M_1	M_2	R	$LO(0,0,1)$	$LO(0,0,2)$
7	all	11	11	7	9	11	11	11
7	mol.	9	9	5	7	9	9	9
8	all	23	21	9	17	21	21	21
8	mol.	18	16	6	13	16	16	16
9	all	47	40	13	22	40	40	40
9	mol.	35	28	7	16	28	28	28
10	all	106	79	18	35	79	79	79
10	mol.	75	49	8	21	49	49	49
11	all	235	152	21	43	152	152	152
11	mol.	159	83	9	25	83	83	83
12	all	551	294	27	57	294	294	294
12	mol.	355	137	10	28	137	137	137

Fixed fractional exponents (exploratory). A script in the repository (directory `structural`) evaluates the nine fixed points

$$(\alpha,\beta) \in \{(0,\tfrac{1}{2}),(\tfrac{1}{2},0),(0,-\tfrac{1}{2}),(-\tfrac{1}{2},0),(\tfrac{1}{2},-1),(-\tfrac{1}{2},1),(\tfrac{1}{3},0),(0,\tfrac{1}{3}),(\tfrac{1}{6},0)\}$$

with $\gamma = 0$ and $\gamma = 1$. We first consider discrimination on all trees of orders 12–16, where every point off the ratio plane with $\gamma = 1$ reaches or nearly reaches the profile floor. At order 16, for example, the counts are 4069, 4067, 4069, 4068, 4069 and 4069 of 4069 profiles for $(0,\frac{1}{2})$, $(\frac{1}{2},0)$, $(0,-\frac{1}{2})$, $(-\frac{1}{2},0)$, $(\frac{1}{3},0)$, $(0,\frac{1}{3})$. With $\gamma = 0$ the same points give 3475, 3940, 3476, 4007, 4009 and 3476, while the two points on the ratio plane give 3996 for both values of γ. We next consider prediction (fixed points, leave-one-out Q^2, no tuning). Here $\gamma = 1$ beats $\gamma = 0$ in 52 of 81 point–dataset combinations, mostly for the vaporisation enthalpy and decane boiling points, and loses mostly on the octane entropy and acentric factor. Every fixed fractional point stays at $Q^2 \leq 0.21$ on the decanes. The full

output is committed with the script.

S6 Structure-sensitivity protocol

The structure-sensitivity criterion for descriptor quality was introduced by Furtula, Gutman and Dehmer [6], and a fingerprint-based variant is due to Rakić and Furtula [17]. We implement the published protocol on the 75 decane isomer trees Ψ. For each tree G, the similar-structure set $S(G) \subset \Psi$ consists of the non-isomorphic trees obtainable from G by relocating a single edge within the molecular class (graph edit distance two, $\Delta \leq 4$ preserved). Then

$$SS(TI, G) = \frac{1}{|S(G)|} \sum_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

$$Abr(TI, G) = \max_{\Omega \in S(G)} \frac{|TI(\Omega) - TI(G)|}{TI(G)},$$

and the class-level $SS(TI)$ and $Abr(TI)$ are the means over all 75 trees. On this class every $S(G)$ is nonempty (sizes 7 to 43). The neighbour relation is symmetric, and $Abr \geq SS$ holds throughout, as it must. As an external control, the implementation reproduces the published decane-class values of Barman and Das [1] for the seven indices common to both studies (M_1, ${}^{m}M_2$, R, χ, H, GA, AG). These give fourteen tabulated SS/Abr values, of which eleven agree exactly at the four displayed decimals and the remaining three differ by one unit in the fourth decimal. The cell-by-cell comparison is in the file `fgd_published_control.csv` of the reproducibility package, while our values for all twelve compared indices are in Table 3 of the main paper. The published desideratum is SS large and Abr small [6]. Under it, the pure-γ point $LO(0, 0, 2)$ has a structure sensitivity indistinguishable at three decimals from that of the hyper-Zagreb index (0.1257 against 0.1256) at a markedly smaller abruptness (0.274 against 0.313). The sensitivity-to-abruptness ratio is our own summary of the two published quantities and is not part of the published criterion. The ratios of the pure-γ points (0.4579 and 0.4565) are comparable to those of $H/2$, GA and R (0.448–0.451). In view of the redundancy analysis of Section 6.1 of

the main paper and Section S1, these third-decimal differences are not a ranking. Two qualifications limit this. First, SS is not a fixed property of the family, since along the pure-γ line it grows with γ (0.031, 0.062, 0.126, 0.319 at $\gamma = 0.5, 1, 2, 5$). The ratio peaks near $\gamma = 2$ and then falls (0.427 at $\gamma = 5$). Second, the parameter-free aggregate $IRLA$, which the pure-γ points approximate to first order in γ, has a higher ratio ($SS = 0.2343$, $Abr = 0.5066$, ratio 0.4625). The favourable sensitivity–abruptness balance is therefore a property of linear imbalance weighting rather than of the exponential form.

S7 Lean formalisation

The directory `formal/` of the repository contains a Lean 4 formalisation (Lean v4.34.1, Mathlib v4.34.1) of the following statements of the main paper. `formal/verify.sh` builds it and checks that no `sorry`, `axiom` or `native_decide` occurs, and that every theorem depends only on the standard axioms (`propext`, `Classical.choice`, `Quot.sound`).

Encoding. Graphs are not formalised as Mathlib graphs, and an edge set is a finite family of edges with a map to (positive real) degree pairs, or a multiset of degree pairs. Hypotheses such as "maximum degree Δ" are stated as bounds on these pairs, so the link "these are the vertex degrees of a graph" is not formalised. The only exception is the two order-13 trees below, which are realised by explicit parent arrays.

Formalised (inequalities and identities, with their hypotheses). The formalised statements are the upper and lower (Δ, δ) bounds, the Cauchy–Schwarz interpolation bound, and the summation step of the ratio and bracket bounds, with the per-edge bounds as hypotheses. They also include the convexity of log-sum-exp for any nonempty index set (via Hölder), which is $\log LO(G;\theta)$ by Eq. (3) of the main paper (that rewriting is not formalised), and the spanning statement of the independence lemma for every $\Delta \geq 3$ (via $\det W = \frac{1}{6}\ln 2\,(8\ln 3 - 13\ln 2) \neq 0$, using $3^8 \neq 2^{13}$). The identity $1 - q^2 = 4xy/(x+y)^2$ is also formalised, as is the order-13 profile pair with equal q-histograms and hence equal pure-γ values for every real γ, together with explicit 13-vertex trees realising both profiles.

The trees are given as parent arrays; that a parent array encodes a tree is a standard fact that is not re-proved in Lean.

Not formalised. The equality characterisations are not formalised, and neither are the gradient and Hessian formulas and strict convexity. The ratio-plane theorem is formalised only through the order-13 pair above; its parts (i) and (ii), the γ-crossing bound and the identifiability proposition are not formalised, as they rest on the Lindemann–Weierstrass theorem, Laguerre's rule and exponential-family theory, none of which is in Mathlib. All statistical and chemometric results are outside the formalisation, as are the closed forms, the extremal-tree proposition and the ceiling and generic-discrimination propositions. The identifiability consequence of Lemma 1 together with its $\Delta = 1, 2$ cases is also not formalised. A statement-by-statement map is the file `LEAN_VERIFICATION.md` of the repository (the encoding is described in `formal/README.md`).

S8 Open provenance items

Every value in Table S5 was traced to a source (per-value record in the repository). The items that remain open are listed here so that they can be read in one place.

(i) *Thirteen gas-phase entropies have no S°(gas) entry in the NIST WebBook.* Six of them (2,4-, 2,5- and 3,3-dimethylhexane, 3-ethylhexane, 2,2,3- and 2,2,4-trimethylpentane) agree, within tolerance or as documented revisions between editions, with the 1947 API Research Project 44 tables or a compiled octane dataset. The remaining seven (4-methylheptane, 2,2-dimethylhexane, 2,3-dimethylhexane, 3,4-dimethylhexane, 3-ethyl-2-methylpentane, 3-ethyl-3-methylpentane, 2,3,3-trimethylpentane) could not be matched to any primary source. The 1947 API Research Project 44 tables [19] list gas-phase entropies for all seven. These differ from the values used by 0.07–2.2 $\mathrm{cal\,mol^{-1}\,K^{-1}}$, as that edition does for isomers whose later values we could match in a compiled octane dataset or the NIST WebBook. No later primary determination was located, so the values used are compiled-dataset values. Substituting

the 1947 values changes the S correlations by at most 0.016 in $|r|$ and puts M_1 ahead of HM by 0.005.

(ii) *2,3,3-trimethylpentane S.* The value 101.31 found in our earlier compilation is a copy of the 2,2,3-trimethylpentane entry and is not used. The value used, 102.06, is that of the compiled 18-octane dataset reproduced by Mondal et al. [13] (the one-decimal compilation of Ediz [5] gives 102.10). This change moves the S correlations by at most 0.009, and the entry used remains a secondary value (item (i)).

(iii) *Four entropies differ from the compiled octane dataset of Ediz [5]* (octane 111.55 vs 111.70; 2,2-dimethylhexane 103.13 vs 103.40; 2,2,4-trimethylpentane 101.81 vs 104.10; 2,3,3-trimethylpentane 102.06 vs 102.10). For octane the value used agrees with the NIST WebBook (111.63). Substituting the compilation values changes all twelve S correlations, by up to 0.043 in $|r|$, and moves the margin between HM and M_1 at the top from 0.001 to 0.007.

(iv) *2,2,3,3-tetramethylbutane ΔH_{vap}.* The compound is crystalline at 298 K (triple point 373.97 K). The benchmark value 8.41 comes from the Milano octane dataset reproduced by Ediz [5], whose ΔH_{vap} column follows the 1947 API Research Project 44 tables [19] at 25 °C for the other seventeen isomers. For this isomer those tables give 10.24, marked as a heat of sublimation, and 7.56 for the liquid at the normal boiling point; 8.41 appears in neither. NIST lists 42.94 and 42.91 kJ mol^{-1} (10.26 kcal mol^{-1}) at 298 K as vaporisation enthalpies, and lists the second determination also as a sublimation enthalpy, so these values describe the crystal. The benchmark value lies just below an estimate for a hypothetical supercooled liquid: the sublimation enthalpy 43.37 minus the fusion enthalpy 7.54 kJ mol^{-1} gives 8.56 kcal mol^{-1}, without a heat-capacity correction. Its original source was not located. The sensitivity of Table 4 of the main paper is reported for each variant separately: original ($|r| = 0.984$ for $LO(0, 0, 1)$), isomer excluded (0.980), hypothetical-liquid estimate (0.976) and sublimation enthalpy (0.374). $LO(0, 0, 1)$ has the largest $|r|$ in all four, although

with the isomer excluded it is level with $LO(0,0,2)$ at three decimals (0.9799 against 0.9795). The matched-budget comparison repeated without the isomer keeps the vaporisation-enthalpy gain in the $(-2,2)$ box (93/100 and 96/100 seeds instead of 100/100). The full trace is the file `TMB_DHVAP_AUDIT.md` of the repository.

(v) *Acentric factors* of 2,2,4-trimethylpentane and 2,2,3,3-tetramethylbutane differ from KDB by 0.002–0.004 (and the latter from the set of [5] by 0.008), and these alternative values change several ω correlations at the third decimal and leave the ω ranking unchanged.

Apart from the sublimation-enthalpy substitution in item (iv), none of these items moves a correlation of Table 4 of the main paper by more than 0.043. The only bolded entry at stake is the S winner, where HM leads M_1 by 0.001 with the values used. The 1947 substitution of item (i) reverses that order by 0.005, a margin without meaning at this resolution. None of the items enters the pre-registered nonane and boiling-point tests of Sections 6.3 and 6.4 of the main paper. Those tests use no entropy, vaporisation-enthalpy or acentric-factor value of Table S5, and the pooled boiling-point test re-extracts the octane boiling points under its own rule. Item (ii) was a transcription error in our earlier data file, corrected here, while the remaining items are limitations of the compiled octane benchmark rather than of the analysis.

Source-quality sensitivity (exploratory). The repository classes every experimental value by the number of independent primary determinations behind it, and agreement with a compilation counts for none. Restricted to values resting on one identified primary study or on two or more research groups, the comparisons of Section 6 of the main paper change as follows. On the decanes (28 of 34 values) the ridge model on degree-pair counts stays ahead of every single index, by 0.856 in Q^2 against $IRLA$, and it is ahead in all 200 refitted subsamples of 80% of the molecules. The restricted pooled set keeps only 42 boiling points, all of nonanes and decanes, because every C6–C8 value is a NIST average. There the ridge model is ahead in 181 of 200 refitted subsamples, so the pooled comparison is not resolved. The registered γ test on the restricted decanes and pooled set again finds no evidence that γ adds value, with LO ahead in 12 and 10 of 50 seeds. No

column leader of Table 4 of the main paper is resolved, and the entropy, acentric-factor and nonane vaporisation-enthalpy comparisons cannot be estimated, since at most five values remain.

Table S5 lists the full 18-isomer dataset used in Section 6 of the main paper, with T_B in °C, ΔH_f and ΔH_{vap} in kcal mol^{-1} (ΔH_{vap} at 298 K), S in cal mol^{-1}K^{-1} and ω dimensionless. The ΔH_{vap} values of 3,3-dimethylhexane (8.97), 3-ethyl-3-methylpentane (9.08), 2,2,3-trimethylpentane (8.83) and 2,3,3-trimethylpentane (8.90) are values consistent with the NIST WebBook determinations, and Section S1 gives the exact-versus-within-uncertainty status of each. The ΔH_{vap} entry of 2,2,3,3-tetramethylbutane is the benchmark-compilation value discussed above. The S value of 2,3,3-trimethylpentane (102.06) is a compiled-dataset value (this section).

Table S5. Physicochemical properties of the 18 octane isomers.

Isomer	T_B	ΔH_f	ΔH_{vap}	S	ω
n-octane	125.6	−49.82	9.92	111.55	0.398
2-methylheptane	117.6	−51.50	9.48	109.84	0.378
3-methylheptane	118.9	−50.82	9.52	111.26	0.371
4-methylheptane	117.7	−50.69	9.48	109.32	0.372
2,2-dimethylhexane	106.8	−53.71	8.92	103.13	0.339
2,3-dimethylhexane	115.6	−51.13	9.27	108.02	0.348
2,4-dimethylhexane	109.4	−52.44	9.03	106.98	0.344
2,5-dimethylhexane	109.1	−53.21	9.05	105.72	0.357
3,3-dimethylhexane	111.9	−52.61	8.97	104.74	0.322
3,4-dimethylhexane	117.7	−50.91	9.32	106.59	0.340
3-ethyl-2-methylpentane	115.6	−50.48	9.21	106.06	0.330
3-ethyl-3-methylpentane	118.3	−51.38	9.08	101.48	0.302
3-ethylhexane	118.5	−50.40	9.48	109.43	0.362
2,2,3-trimethylpentane	109.8	−52.61	8.83	101.31	0.300
2,2,4-trimethylpentane	99.2	−53.57	8.40	101.81	0.305
2,3,3-trimethylpentane	114.8	−51.73	8.90	102.06	0.291
2,3,4-trimethylpentane	113.5	−51.97	9.01	102.39	0.317
2,2,3,3-tetramethylbutane	106.5	−53.99	8.41	93.06	0.247

S9 Additional tables for Section 6 of the main paper

Reporting conventions and the boiling-point extraction rule

In Table 5 of the main paper and in Tables S6 and S7 medians are taken over candidate-stream seeds, and seed counts describe the random search on fixed data rather than replications. Values are rounded once for display, so a tabulated difference may differ in the last place from the difference of two displayed entries. In Table S6 the narrow- and wide-box streams use the same uniform variates scaled by the half-width, so comparisons across boxes are paired there; the boiling-point tests of Table 5 of the main paper draw their candidate streams sequentially.

The extraction rule of Section 6.4 of the main paper takes the NIST average, or else the median of non-compilation determinations if they agree within 3 K, and otherwise excludes the molecule. This gives all hexanes, heptanes and octanes and 34 of the 35 nonanes, since one nonane has two NIST entries with the same InChI whose boiling points differ by 6.4 K. It also gives 34 of the 75 decanes, because 31 of them have no NIST boiling point and 10 have determinations more than 3 K apart. For 14 of the 34 included decanes NIST lists a single non-compilation determination, so the agreement rule does not apply to them. NIST averages, 57 of the 100 values, are used as given, and 1 further value is the mean of two NIST averages that agree within 3 K, so 58 values rest on NIST averages. Neither the quoted uncertainty of an average (up to 6 K) nor the spread of its constituent points is screened. Without the two averages quoting more than 3 K the decane ridge gives $Q^2 = 0.863$ (exploratory). Under this uniform rule the pooled octane boiling points can differ from those of Table S5 by up to 0.5 °C, for example 2,4-dimethylhexane at 108.9 against 109.4 °C, a documented source variation. Seven pooled nonane boiling points differ from the replication-set values of Section 6.3 of the main paper by 0.1–0.9 °C, as that protocol took the most recent determination where NIST gives no average. The included and excluded decanes do not

differ detectably in branching or imbalance, measured by the numbers of tertiary and quaternary carbons and of methyl groups, the Wiener index, $IRLA$, $LO(0,0,1)$ and R. Two-sided permutation tests gave $p \geq 0.10$ in every case, though with 34 and 41 molecules such tests can rule out only large differences.

Table S6 is the search-box sensitivity of the matched comparison of Section 6.2 of the main paper. Edge is the share of outer folds, pooled over the 50 seeds, in which the parent's selection lies within 10% of the boundary of the box of the stated half-width. Medians of Q^2 are over seeds 0–49 at 2000 candidates per model. The counts are the number of seeds in which LO beats GM (half-width 2) or LO_h beats GM (half-width 12), where LO_h is the hybrid control with (α, β) in the wide box and $\gamma \in (-2, 2)$. Table S7 gives the within-fold baselines of the nonane replication of Section 6.3 of the main paper. Its Class. column is the best of the ten classical reductions chosen within each fold; the pre-registered fixed-index baseline of the protocol used all twelve compared indices, including the two pure-γ points, and is reported in the repository.

Table S6. Sensitivity of the matched comparison to the search box (medians of Q^2 over seeds 0–49 at 2000 candidates per model).

	half-width 2				half-width 12				
Set, prop.	Edge	GM	LO	>	Edge	GM	LO	LO_h	>
oct. T_B	69%	0.665	0.730	36	2%	0.606	0.495	0.544	21
oct. ΔH_f	52%	0.821	0.769	11	0%	0.820	0.670	0.761	10
oct. ΔH_{vap}	97%	0.887	0.957	50	3%	0.942	0.934	0.946	28
oct. S	85%	0.904	0.891	11	4%	0.842	0.838	0.867	38
oct. ω	0%	0.979	0.974	10	0%	0.979	0.938	0.951	7
non. T_B	61%	0.576	0.601	30	0%	0.524	0.392	0.577	31
non. ΔH_{vap}	93%	0.855	0.895	49	9%	0.892	0.881	0.889	24

S10 Further results on the Loyola family

This section holds material moved from the main paper to shorten it: the proofs of Lemma 1 and Theorem 1 of the main paper, per-graph iden-

Table S7. Nested leave-one-out Q^2 on the octanes (oct., $n = 18$) and the nonane replication set (non., $n = 35$ for T_B, 34 for ΔH_{vap}). Mean: training-fold mean. Class.: best of the ten classical indices, selected inside each fold. IRLA: one-descriptor model on $IRLA$. Ridge: ridge regression on the edge-degree-pair counts. LO_1: the fixed, untuned point $LO(0, 0, 1)$. GM, LO: median of Q^2 over seeds 0–99 at budget 200 in the box of half-width 2 (see Table S6 for wider boxes). Last column: seeds in which LO beat GM at budgets 200/500.

Set, property	Mean	Class.	IRLA	Ridge	LO_1	GM	LO	LO>GM
oct. T_B	−0.121	0.593	0.585	0.344	0.591	0.662	0.652	53/64
oct. ΔH_f	−0.121	0.746	0.593	0.654	0.639	0.831	0.689	14/24
oct. ΔH_{vap}	−0.121	0.884	0.948	0.954	0.957	0.888	0.953	100/100
oct. S	−0.121	0.905	0.733	0.721	0.785	0.904	0.889	34/31
oct. ω	−0.121	0.971	0.810	0.851	0.823	0.979	0.970	26/28
non. T_B	−0.060	0.502	0.400	0.682	0.409	0.577	0.586	58/63
non. ΔH_{vap}	−0.062	0.857	0.897	0.927	0.903	0.853	0.900	95/98

tifiability, the limits along the γ axis, closed forms on standard graphs and termwise bounds. Statements and proofs are unchanged apart from cross-references. Numbers with the prefix S refer to this supplement.

S10.1 Proofs of Lemma 1 and Theorem 1 of the main paper

Proof of Lemma 1 of the main paper. The vectors $(1, \ln(ij), \ln(i + j))$ on the pairs $(1, 1), (2, 2), (1, 2)$ have rank 3, since the corresponding 3×3 determinant equals $\ln 2 \cdot \ln(9/8) \neq 0$, so it suffices to show that q_{ij} is not an affine function of $\big(\ln(ij), \ln(i + j)\big)$ on P_Δ. The four pairs $(1, 1), (2, 2), (1, 2), (1, 3)$, all admissible once $\Delta \geq 3$, witness this. The pair $(1, 1)$ occurs only in K_2, and the conclusion persists without it, since the augmented vectors of $(1, 2), (1, 3), (2, 2), (3, 3)$, namely $(1, \ln 2, \ln 3, \frac{1}{3})$, $(1, \ln 3, \ln 4, \frac{1}{2})$, $(1, \ln 4, \ln 4, 0)$ and $(1, \ln 9, \ln 6, 0)$, have determinant $-\frac{1}{6} \ln \frac{3}{2} \ln\big(2^{13}/3^8\big) \approx -0.0150 \neq 0$. Suppose $q_{ij} = a + b \ln(ij) + c \ln(i + j)$ on these pairs. The pairs $(1, 1)$ and $(2, 2)$ have $q = 0$, so they give $a + c \ln 2 = 0$ and $a + (2b + 2c) \ln 2 = 0$, hence $a = -c \ln 2$ and $b = -c/2$. Substituting into the $(1, 2)$ and $(1, 3)$ equations gives $\frac{1}{3} = c\,(\ln 3 - \frac{3}{2} \ln 2)$

and $\frac{1}{2} = c\,(\ln 2 - \frac{1}{2}\ln 3)$. In particular $c \neq 0$, since the left side $\frac{1}{3}$ is nonzero. Three times the first minus twice the second gives $0 = c\,(4\ln 3 - \frac{13}{2}\ln 2)$, hence $8\ln 3 = 13\ln 2$, i.e. $3^8 = 2^{13}$. This is false ($6561 \neq 8192$). Hence the augmented vectors have rank 4. Indeed, a linear relation among the four coordinates on P_Δ with a nonzero coefficient on q_{ij} would express q_{ij} as an affine function of $\ln(ij)$ and $\ln(i+j)$, which we have just excluded. A relation with zero coefficient on q_{ij} is excluded by the rank-3 determinant. If $e^{\theta^\top z} = c\,e^{\theta'^\top z}$ on P_Δ, then $(\theta - \theta')^\top z_{ij} = \ln c$ for all admissible pairs, so $(\ln c, \theta' - \theta)$ is orthogonal to every augmented vector. Since the augmented vectors span $\mathbb{R}^4$, the only vector orthogonal to all of them is 0, so $\theta = \theta'$ and $c = 1$. For $\Delta = 2$ only the pairs $(1,1), (1,2), (2,2)$ exist. The 3×3 determinant computed at the start of the proof shows that their three augmented vectors have rank 3, and three vectors cannot have more. For $\Delta = 1$ the single augmented vector has rank 1. ■

Proof of Theorem 1 of the main paper. Direct differentiation of $LO_G(\theta) = \sum_{uv} e^{\theta^\top z_{uv}}$ gives $\partial_k F_G = \sum_{uv} w_{uv} z_{uv,k} = \mathbb{E}_\theta[z_k]$ and $\partial_k \partial_l F_G = \mathbb{E}_\theta[z_k z_l] - \mathbb{E}_\theta[z_k]\,\mathbb{E}_\theta[z_l]$, the (k,l)-entry of $\mathrm{Cov}_\theta(z)$, which is positive semidefinite. Convexity follows. The measure $w(\theta)$ has full support on E for every θ. Hence $\mathrm{Cov}_\theta(z)$ is singular in a direction v iff $v^\top z_{uv}$ is constant over all edges, i.e. iff the z_{uv} lie in a proper affine subspace. If they do lie in such a subspace, then along the corresponding v the function $t \mapsto F_G(\theta + tv)$ is affine, so F_G is not strictly convex. If they do not, the Hessian is positive definite at every θ, which implies strict convexity. This proves the strictness criterion. Since $q_{uv} \in [0,1)$, $\mathbb{E}_\theta[q] \in [0,1)$, and $\mathbb{E}_\theta[q] = 0$ iff $q_{uv} = 0$ on every edge. On an r-regular graph all z_{uv} equal $z_0 = (2\ln r, \ln 2r, 0)$. ■

S10.2 Per-graph identifiability and limits along γ

Remark (Independence versus per-graph identifiability). Lemma 1 of the main paper concerns the edge-weight *function* on the full admissible pair set. On a single graph G the index can merely rescale along a direction v, $LO(G;\theta + tv) = e^{ct}\,LO(G;\theta)$, exactly when the feature vectors z_{ij} of its edges fail to affinely span $\mathbb{R}^3$ (Proposition S1). On an r-regular graph,

for instance, $LO(G;\theta) = m\, e^{\theta^\top z_0}$ with $z_0 = (2\ln r, \ln 2r, 0)$, so the values determine only $\theta^\top z_0$. The lemma guarantees that no such degeneracy is forced once all pairs with $\Delta \geq 3$ are available.

For a graph G let $R(G)$ be the set of degree pairs it realises, $r(G)$ the number of distinct imbalances among them, and $A(G)$ the affine dimension of $\{z_{ij} : (i,j) \in R(G)\}$, that is, the dimension of the smallest affine subspace of $\mathbb{R}^3$ containing it.

Proposition S1 (Per-graph identifiability)**.** *(i) $A(G)$ equals the rank of the Hessian $\mathrm{Cov}_\theta(z)$ of $\log LO_G$ at every θ, and the directions v for which there is a constant $c = c(v)$ with $LO(G;\theta + tv) = e^{ct}\, LO(G;\theta)$ for all t and θ form a subspace of dimension $3 - A(G)$.*

(ii) $A(G) \leq \min\big(|R(G)| - 1,\ r(G),\ 3\big)$ for every graph.

(iii) If $r(G) = 2$ with imbalances $q_1 \neq q_2$, then $v = (1, -2, c)$ with $c = [\ln(1-q_2^2) - \ln(1-q_1^2)]/(q_1 - q_2)$ is such a direction.

Proof. (i) The measure $w(\theta)$ puts positive weight on every edge, so, as in the proof of Theorem 1 of the main paper, $\mathrm{Cov}_\theta(z)$ is singular exactly in the directions v with $v^\top z$ constant on $R(G)$. Such v are those orthogonal to the differences of the z_{ij}, and they form a subspace of dimension $3 - A(G)$, so the rank of $\mathrm{Cov}_\theta(z)$ is $A(G)$. (In the language of exponential families [2], $w(\theta)$ is the family with sufficient statistic z on the support $R(G)$. The rank statement says that its minimal representation has $A(G)$ parameters.) A direction v rescales the index, $LO(G;\theta + tv) = e^{ct} LO(G;\theta)$ for all θ and t, exactly when $v^\top \nabla F_G(\theta) = c$ is independent of θ. Differentiating, this holds exactly when $\mathrm{Cov}_\theta(z)\, v = 0$ for all θ, that is, when $v^\top z$ is constant on $R(G)$. The rescaling directions are therefore the subspace of dimension $3 - A(G)$ just described. (ii) By Eq. (4) of the main paper, $z_{ij} = (2s + \ln(1-q^2) - \ln 4,\ s,\ q)$ with $s = \ln(i+j)$, so pairs of equal imbalance lie on one line parallel to $(2,1,0)$, and r parallel lines span an affine space of dimension at most r. (iii) $v^\top z_{ij} = \ln(1-q_{ij}^2) - \ln 4 + c\, q_{ij}$ takes the same value at q_1 and q_2 by the choice of c. ∎

Remark (The case $\Delta = 7$). Equality holds in (ii) for every set of admissible pairs with $\Delta \leq 6$ (by exhaustive enumeration), hence for all molecular

graphs. It fails from $\Delta = 7$: among the subsets of P_7 with at most four pairs, the only one violating equality is $\{(1,7),(2,6),(3,5),(4,4)\}$, whose degree sums all equal 8. Every violating set contains a violating subset of at most four pairs. If $A(R) = a$ is below the bound, then $a \le 2$, and R has at least $a+1$ imbalances and $a+2$ pairs. One pair from each of $a+1$ imbalance classes and one further pair form a subset S of $a+2 \le 4$ pairs, with $A(S) \le a$ while its bound is $a+1$. Such a subset consists of three collinear points with two imbalances or four coplanar points with three imbalances. Any fifth pair of P_7 has a degree sum different from 8 and leaves that plane, so this is the only violating set. A constant degree sum makes $\ln(i+j)$ constant on the set, so $A \le 2$, while the bound equals 3 as soon as the set has at least four pairs and three imbalances. In a connected graph whose edges all have degree sum s, each neighbour of a vertex of degree d has degree $s-d$, so only the pair $\{d, s-d\}$ occurs. Hence no connected graph realises this set.

Example S1. The spider with four legs of length three (a centre vertex joined to four paths on three vertices) realises $(2,4)$, $(2,2)$ and $(1,2)$, with the two imbalances $\frac{1}{3}$ and 0. On it γ is confounded with (α,β) along the direction of part (iii).

Proposition S2 (Limits along the γ axis)**.** *Let G have at least one edge, let $E_{\max}(G)$ denote the set of edges realising* $\mathrm{imb}_{\max}(G) = \max_{uv} \mathrm{imb}(d_u, d_v)$ *and $E_{\min}(G)$ the set realising* $\mathrm{imb}_{\min}(G) = \min_{uv} \mathrm{imb}(d_u, d_v)$*. Then as $\gamma \to +\infty$,*

$$\frac{LO(G;\alpha,\beta,\gamma)}{e^{\gamma\,\mathrm{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^{\alpha} (d_u + d_v)^{\beta},$$

and as $\gamma \to -\infty$,

$$LO(G;\alpha,\beta,\gamma)/e^{\gamma\,\mathrm{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^{\alpha} (d_u + d_v)^{\beta}.$$

Proof. With $\mu(G) := \mathrm{imb}_{\max}(G)$ and $\varepsilon_{uv} := \mu(G) - \mathrm{imb}(d_u, d_v) \ge 0$, we have $LO(G;\alpha,\beta,\gamma)/e^{\gamma\mu(G)} = \sum_{uv} (d_u d_v)^{\alpha} (d_u + d_v)^{\beta} e^{-\gamma\varepsilon_{uv}}$. The summands over $E_{\max}(G)$ ($\varepsilon_{uv} = 0$) are constant, and the finitely many others

tend to 0 as $\gamma \to +\infty$. The case $\gamma \to -\infty$ is dual. ■

S10.3 Closed-form values on standard graphs

The classical graph families used are the complete graph K_n, the cycle C_n, the path P_n, the star S_n and the complete bipartite graph $K_{a,b}$. Beyond these, the hypercube Q_k has 2^k vertices and $k \cdot 2^{k-1}$ edges and is k-regular. The wheel W_n is an $(n-1)$-cycle plus a universal vertex, and the friendship graph F_p consists of p triangles sharing a common vertex. We also treat r-regular graphs.

Proposition S3 (Closed forms)**.** *Let $\alpha, \beta, \gamma \in \mathbb{R}$. Then:*

1. $LO(K_n; \alpha, \beta, \gamma) = \dfrac{n(n-1)}{2}\, 2^{\beta} (n-1)^{2\alpha+\beta}$ *for* $n \geq 2$.
2. $LO(C_n; \alpha, \beta, \gamma) = n \cdot 4^{\alpha+\beta}$ *for* $n \geq 3$.
3. $$LO(P_n; \alpha, \beta, \gamma) = \begin{cases} 2^{\beta}, & n = 2, \\ 2 \cdot 2^{\alpha} 3^{\beta} e^{\gamma/3}, & n = 3, \\ 2 \cdot 2^{\alpha} 3^{\beta} e^{\gamma/3} + (n-3) \cdot 4^{\alpha+\beta}, & n \geq 4. \end{cases}$$
4. $LO(S_n; \alpha, \beta, \gamma) = (n-1)\,(n-1)^{\alpha} n^{\beta} \exp\!\left(\gamma\, \dfrac{n-2}{n}\right)$ *for* $n \geq 2$.
5. $LO(K_{a,b}; \alpha, \beta, \gamma) = ab\,(ab)^{\alpha} (a+b)^{\beta} \exp\!\left(\gamma\, \dfrac{|a-b|}{a+b}\right)$ *for* $a, b \geq 1$.
6. $LO(Q_k; \alpha, \beta, \gamma) = k \cdot 2^{k-1} \cdot 2^{\beta} k^{2\alpha+\beta}$ *for* $k \geq 1$.
7. *For $n \geq 5$, W_n has $n-1$ spoke edges with degree pair $(3, n-1)$ and $n-1$ rim edges with degree pair $(3,3)$, so*
$$LO(W_n; \alpha, \beta, \gamma) = (n-1)\Big[\, 3^{\alpha} (n-1)^{\alpha} (n+2)^{\beta} \exp\!\Big(\gamma\, \tfrac{n-4}{n+2}\Big) + 9^{\alpha} 6^{\beta} \Big].$$
8. *For $p \geq 1$, F_p has $2p$ spoke edges $(2, 2p)$ and p outer edges $(2,2)$, so*
$$LO(F_p; \alpha, \beta, \gamma) = p \cdot 4^{\alpha+\beta} + 2p\,(4p)^{\alpha} (2p+2)^{\beta} \exp\!\Big(\gamma\, \tfrac{p-1}{p+1}\Big).$$
9. *For an r-regular graph G with $r \geq 1$ and m edges, $LO(G; \alpha, \beta, \gamma) = m \cdot 2^{\beta} r^{2\alpha+\beta}$.*

Proof. Each case follows by enumerating the edge degree pairs and applying Definition 1 of the main paper. In K_n, C_n, Q_k and r-regular graphs every edge has $d_u = d_v = r$, so imb $= 0$ and $(d_u d_v)^\alpha (d_u + d_v)^\beta = 2^\beta r^{2\alpha+\beta}$, which gives items (1), (2), (6) and (9). All $n-1$ edges of S_n have degree pair $(1, n-1)$ with imb $= (n-2)/n$ (item (4)), and all ab edges of $K_{a,b}$ have pair (a, b) (item (5)). For $n \geq 4$, P_n has two pendant edges $(1, 2)$ and $n-3$ interior edges $(2, 2)$, and the cases $n = 2, 3$ are immediate (item (3)). Items (7) and (8) follow by summing over the two edge-degree-pair classes of W_n and F_p. ∎

S10.4 Termwise bounds

Throughout this subsection G is a simple connected graph with $\delta \leq d_v \leq \Delta$ for every $v \in V(G)$. For $\gamma \geq 0$ every edge satisfies $1 \leq e^{\gamma q_{uv}} \leq e^{\gamma(\Delta-1)/(\Delta+1)}$. At $\gamma = 0$ bounds of the types below, with their equality cases, are known [7, 12, 16], and inequalities between exponential indices are given in [3]. The statements below extend them to $\gamma \neq 0$ (Propositions S5 and S6 to $\gamma > 0$ only) and track how γ changes the equality cases. The inequalities of the Cauchy–Schwarz interpolation corollary (Section S10.4) and of Propositions S5 and S6, not their equality cases, are machine-checked. For Propositions S4 and S7 only the summation step is, with the edgewise extrema taken as hypotheses (Section S7).

Definition S1 (Edge weight)**.** For real parameters α, β, γ and positive reals x, y,

$$\psi(x, y; \alpha, \beta, \gamma) = (xy)^\alpha (x + y)^\beta \exp\!\left(\gamma \, \tfrac{|x-y|}{x+y}\right). \tag{S1}$$

With this notation $LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} \psi(d_u, d_v; \alpha, \beta, \gamma)$.

Proposition S4 (Bracket bound)**.** *Let G be a simple connected graph of size $m \geq 1$ with degrees in $[\delta, \Delta]$. Then*

$$m \cdot \psi_{\min}(\delta, \Delta; \alpha, \beta, \gamma) \;\leq\; LO(G; \alpha, \beta, \gamma) \;\leq\; m \cdot \psi_{\max}(\delta, \Delta; \alpha, \beta, \gamma), \tag{S2}$$

where $\psi_{\min} = \min\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$ and $\psi_{\max} = \max\{\psi(i, j; \alpha, \beta, \gamma) : \delta \leq i \leq j \leq \Delta\}$. The upper inequality is an equality

if and only if every edge of G has a degree pair attaining $\psi_{\max}$. The lower inequality is an equality if and only if every edge has a degree pair attaining $\psi_{\min}$.

Proof. Every edge uv has its degree pair in the admissible set $\mathcal{P}(\delta,\Delta) := \{(i,j) \in \mathbb{Z}^2 : \delta \le i \le j \le \Delta\}$, hence $\psi_{\min} \le \psi(d_u, d_v; \alpha, \beta, \gamma) \le \psi_{\max}$ on every edge, and summing over the m edges and using $\sum_{uv} \psi(d_u, d_v; \alpha, \beta, \gamma) = LO(G; \alpha, \beta, \gamma)$ gives (S2). Each side of (S2) is the sum of m per-edge inequalities, and is therefore an equality if and only if every per-edge inequality is. That is, equality holds if and only if every edge has a degree pair attaining $\psi_{\max}$ (upper side), respectively $\psi_{\min}$ (lower side), over $\mathcal{P}(\delta,\Delta)$. ∎

Corollary (Cauchy–Schwarz interpolation)**.** *Let $\alpha_1 + \alpha_2 = 2\alpha$, $\beta_1 + \beta_2 = 2\beta$, and $\gamma_1 + \gamma_2 = 2\gamma$. Then for every simple connected graph G with $m \ge 1$ edges,*

$$LO(G;\alpha,\beta,\gamma) \;\le\; \sqrt{LO(G;\alpha_1,\beta_1,\gamma_1)\cdot LO(G;\alpha_2,\beta_2,\gamma_2)}$$

Equality holds if and only if the ratio

$$\psi(d_u, d_v; \alpha_1, \beta_1, \gamma_1) / \psi(d_u, d_v; \alpha_2, \beta_2, \gamma_2)$$

is the same for every edge of G.

Proof. For each edge $e = uv$ write $\psi_e^{(k)} := \psi(d_u, d_v; \alpha_k, \beta_k, \gamma_k) > 0$, $k \in \{1,2\}$, and by additivity of the exponents and the hypotheses on $(\alpha_k, \beta_k, \gamma_k)$, $\psi_e^{(1)}\psi_e^{(2)} = \psi(d_u, d_v; \alpha, \beta, \gamma)^2$ on every edge. The Cauchy–Schwarz inequality for $a_e = (\psi_e^{(1)})^{1/2}$ and $b_e = (\psi_e^{(2)})^{1/2}$ gives $\sum_e a_e b_e \le (\sum_e a_e^2)^{1/2} (\sum_e b_e^2)^{1/2}$, which is the claim because $\sum_e a_e b_e = LO(G;\alpha,\beta,\gamma)$, $\sum_e a_e^2 = LO(G;\alpha_1,\beta_1,\gamma_1)$ and $\sum_e b_e^2 = LO(G;\alpha_2,\beta_2,\gamma_2)$. Equality holds if and only if $b_e = \lambda a_e$ for a constant $\lambda \ge 0$ and all e, i.e. if and only if $\psi_e^{(2)}/\psi_e^{(1)} = \lambda^2$ is the same on every edge. (The inequality itself is midpoint log-convexity of $\theta \mapsto LO(G;\theta)$, Theorem 1 of the main paper.) ∎

Proposition S5 (Upper bound in m and Δ)**.** *Let G be a connected graph of size $m \geq 1$ and maximum degree Δ. For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \;\leq\; m \cdot \Delta^{2\alpha} \cdot (2\Delta)^{\beta} \cdot \exp\!\left(\gamma\, \frac{\Delta - 1}{\Delta + 1}\right). \tag{S3}$$

For $\Delta = 1$ the graph is K_2 and the bound is attained for every (α, β, γ). For $\Delta \geq 2$, equality holds if and only if $\alpha = \beta = \gamma = 0$, or $\gamma = 0$, $\alpha + \beta > 0$ and G is Δ-regular, or $\gamma > 0$, $\alpha = \beta = 0$ and $G \cong K_{1,\Delta}$; for $\Delta \geq 2$, $\gamma > 0$ and $\alpha + \beta > 0$ the inequality is strict.

Proof. On every edge uv we have $1 \leq d_u, d_v \leq \Delta$, hence $d_u d_v \leq \Delta^2$ and $d_u + d_v \leq 2\Delta$, and the normalised imbalance $\mathrm{imb}(x, y) = |x - y|/(x + y)$ is maximised over $\{(i, j) : 1 \leq i \leq j \leq \Delta\}$ at $(1, \Delta)$, so $|d_u - d_v|/(d_u + d_v) \leq (\Delta - 1)/(\Delta + 1)$. Since $\alpha, \beta, \gamma \geq 0$, the maps $t \mapsto t^{\alpha}$, $t \mapsto t^{\beta}$ and $t \mapsto e^{\gamma t}$ are non-decreasing, so $\psi(d_u, d_v; \alpha, \beta, \gamma) \leq \Delta^{2\alpha}(2\Delta)^{\beta} \exp(\gamma(\Delta - 1)/(\Delta + 1))$, and summing over the m edges yields (S3). For $\gamma = 0$ the exponential factor is 1. If $\alpha = \beta = 0$ both sides equal m, and if $\alpha + \beta > 0$ equality forces $d_u d_v = \Delta^2$ or $d_u + d_v = 2\Delta$ on every edge, that is $d_u = d_v = \Delta$, so G is Δ-regular; the converse is immediate. For $\gamma > 0$ and $\Delta \geq 2$ the bound is attained if and only if $\alpha = \beta = 0$ and every edge has degree pair $(1, \Delta)$ (for connected G, $G \cong K_{1,\Delta}$). If $\alpha > 0$ or $\beta > 0$ the three factor-wise maxima are not simultaneously attainable and the inequality is strict. ∎

Proposition S6 (Lower bound in m and δ)**.** *Let G be a connected graph of size $m \geq 1$ and minimum degree δ. For $\alpha, \beta \geq 0$ and $\gamma \geq 0$,*

$$LO(G; \alpha, \beta, \gamma) \;\geq\; m \cdot \delta^{2\alpha} \cdot (2\delta)^{\beta}, \tag{S4}$$

with equality if and only if G is δ-regular, except in the degenerate case $(\alpha, \beta, \gamma) = (0, 0, 0)$, where $LO(G; \alpha, \beta, \gamma) \equiv m$ and the bound holds with equality for every graph.

Proof. On every edge $d_u, d_v \geq \delta$, so for $\alpha, \beta \geq 0$, $(d_u d_v)^{\alpha} \geq \delta^{2\alpha}$ and $(d_u + d_v)^{\beta} \geq (2\delta)^{\beta}$. For $\gamma \geq 0$ the exponential factor is at least 1 because the imbalance is non-negative, and multiplying and summing over the m edges yields (S4). If $(\alpha, \beta, \gamma) = (0, 0, 0)$ then $LO(G; 0, 0, 0) = m$ equals the

bound for every graph, while otherwise equality requires equality in every active per-edge factor, which for $\alpha > 0$ or $\beta > 0$ forces $d_u = d_v = \delta$ on every edge. If $\alpha = \beta = 0$ and $\gamma > 0$ it forces $d_u = d_v$ on every edge, so the connected graph G is regular, hence δ-regular. Conversely, on a δ-regular graph every edge has degree pair (δ, δ) and equality holds. ■

The next proposition gives a single uniform template for bounding $LO(G; \alpha, \beta, \gamma)$ both above and below. The bound is in terms of any classical bond-incident-degree index whose edge contribution is strictly positive on the admissible degree pairs of the graph.

Proposition S7 (Ratio bound)**.** *Let G be a simple graph with $E(G) \neq \varnothing$ and maximum degree at most Δ. Let $I_h(G) = \sum_{uv \in E(G)} h(d_u, d_v)$ be a BID index with edge contribution $h(i, j)$ strictly positive on every degree pair (i, j) that appears in G. Define*

$$c^h_{\min} = \min_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)}, \qquad c^h_{\max} = \max_{(i,j)} \frac{\psi(i, j; \alpha, \beta, \gamma)}{h(i, j)},$$

where the extrema range over the degree pairs actually appearing in G. Then

$$c^h_{\min} \cdot I_h(G) \;\leq\; LO(G; \alpha, \beta, \gamma) \;\leq\; c^h_{\max} \cdot I_h(G), \tag{S5}$$

with equality on either side if and only if every edge of G attains the corresponding extremum of ψ/h.

Proof. On every edge uv we have $h(d_u, d_v) > 0$ and $c^h_{\min} \leq \psi(d_u, d_v; \alpha, \beta, \gamma)/h(d_u, d_v) \leq c^h_{\max}$. Multiplying by $h(d_u, d_v)$ and summing over $E(G)$, with $\sum_{uv} h(d_u, d_v) = I_h(G)$ and $\sum_{uv} \psi = LO(G; \alpha, \beta, \gamma)$, gives (S5). Each side is an equality if and only if every edge attains the corresponding extremum of ψ/h over the appearing pairs. ■

Proposition S7 is a *graph-specific* comparison, whose constants depend on the degree pairs that actually appear in G. Its constants are the best edgewise ones, but a bound is attained only when every edge attains the extremum. A uniform version follows by enlarging the range of the extrema.

Corollary (Uniform ratio bound)**.** *Assume the hypotheses of Proposition S7, but with h strictly positive on all of $P_\Delta = \{(i,j) : 1 \le i \le j \le \Delta\}$ and the extrema taken over P_Δ. Then inequality* (S5) *holds simultaneously for every graph of maximum degree at most Δ, with constants independent of the individual graph. Equality on either side holds if and only if every edge of G realises a degree pair attaining the corresponding extremum over P_Δ. The uniform constants are never better than the graph-specific ones, and are strictly worse unless G contains a degree pair attaining the corresponding extremum over P_Δ.*

S11 Supporting analyses for Section 6 of the main paper

This section holds the parts of Sections 6.2–6.4 of the main paper that were moved here to shorten it, together with the list of changes made to the pre-registered protocols after outcomes had been seen (Table S8). The text is unchanged apart from cross-references.

S11.1 Search box (Section 6.2 of the main paper)

On a dense parent grid (steps 0.05, 0.1, 0.2 and 0.25) the nested Q^2 of the parent for octane ΔH_{vap} is 0.8873, 0.9156, 0.9525 and 0.9383 at half-widths 2, 4, 8 and 12. The selected point lies within one grid step of the boundary in 18, 18, 18 and 0 of the 18 folds. On the nonanes the values are 0.8532, 0.8779, 0.8997 and 0.8837, with 34, 34, 34 and 0 boundary folds of 34. The half-width-12 box, chosen after inspecting these curves as the smallest examined in which the parent's selections are interior, is used below for every property. This box does not favour the parent uniformly. For T_B it lowers the parent's nested Q^2 from 0.665 to 0.606 on the octanes and from 0.576 to 0.524 on the nonanes.

The fixed point $LO(0,0,1)$, singled out on these octanes, and the parameter-free $IRLA$ match the fixed parent point $M_{4.5,-9}$, which lies near the parent's wide-box selections (leave-one-out $Q^2 = 0.957$ and 0.948 against 0.949, on the octanes). That point was read off those selections, so

its Q^2 is post-selection; the nested wide-box parent gives 0.938 to 0.942.

S11.2 Untuned point, $IRLA$ and ridge on the nonanes (Section 6.3 of the main paper)

On both sets the untuned point $LO(0, 0, 1)$ matches or beats the tuned Loyola model in point estimate (0.957 and 0.903 against medians 0.953 and 0.900); on the octanes this point was singled out from the same data (Table 4 of the main paper). Since $LO(G; 0, 0, \gamma) = m + \frac{\gamma}{2}\, IRLA(G) + O(\gamma^2)$ as $\gamma \to 0$, it is closely related to $IRLA$; the correlation between the two is 0.996 on both sets, and $IRLA$ comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). A term linear in the imbalance suffices for this property in point estimate, but it is not necessary: the wide-box parent, which has no such term, comes close through $(1 - q^2)^\alpha$ (Section 6.2 of the main paper).

Ridge regression on the degree-pair counts, a reference for what the counts carry, has a higher point estimate than every index model on the nonanes ($Q^2 = 0.927$ for ΔH_{vap}, 0.682 for T_B), although the conditional interval for LO minus ridge on nonane ΔH_{vap}, $[-0.101, +0.032]$, includes zero. On the octanes ridge is below the best index model for four of the five properties and level on ΔH_{vap} (0.954 against 0.957 for $LO(0, 0, 1)$; Table S7).

S11.3 Boiling points (Section 6.4 of the main paper)

In the original box of half-width 2 the two tuned models are level on the decanes, and the Loyola model is marginally ahead on the pooled set (Section 6.4 of the main paper). Seed by seed, that narrow-box Loyola model beats the wide-box parent GM in 38 of 50 seeds on the pooled set (median seed-matched difference +0.0033) but in only 22 of 50 on the decanes (median −0.037). Against the narrow-box parent the counts are 40 and 25 of 50. These comparisons were not registered, carry no interval and count optimiser variance on fixed data, so they do not indicate a gain from γ.

The pooled cohort was corrected after an external audit (Table S8).

On the uncorrected cohort of 101 molecules the same test gave -0.0001, 24 of 50, $[-0.003, +0.002]$, so the correction changes no conclusion. The intervals are conditional on the fitted predictions and omit refitting and selection.

Wiener's two-descriptor model [20], with the path number W and Wiener's polarity number p_3, the number of carbon pairs three bonds apart, fitted without tuning, gives $Q^2 = 0.904$ on the decanes. On trees $p_3 = \sum_{uv}(d_u - 1)(d_v - 1)$, since a path of length three is fixed by its middle edge, so p_3 is a linear function of the degree-pair counts and the distance information in this model is W. This single point estimate, given without an interval and for descriptors historically chosen on alkane boiling points, suggests that within one carbon number W may add to the degree-pair information; its difference from ridge is not resolved here.

S11.4 Changes to the pre-registered protocols

Table S8. Changes to the pre-registered protocols made after outcomes had been seen. Protocols and change records are in the reproducibility package [15].

Test	Change after outcomes	Effect
Nonanes (Section 6.3 of the main paper)	Parent box widened from half-width 2 to 12, chosen after inspecting the ΔH_{vap} curves	Registered verdict reported as obtained; the gain shrinks to about zero in the wide box
Nonanes	Seed-count criterion judged to count optimiser variance	None on reported numbers
Nonanes	Measured boiling points made the primary predictive evidence	None on reported numbers
Boiling points, Test B	All NIST pages of a skeleton pooled; one nonane excluded, one heptane value moved by 0.2 °C	Verdict unchanged; both cohorts reported
Nonanes	A second NIST T_B of 3-ethyl-4-methylhexane, 6.4 K lower, found later	Still mixed if excluded or replaced (Section S4)
Boiling points, Test B	Least-squares solver rescaled by columns	Verdict unchanged
Boiling points, Test B	Scale-invariant rank test	No displayed value changes
Boiling points, Test B	Stricter rank threshold and a deletion-rank guard in the size-adjusted scorer	No output changes
Boiling points, both	Interval rule read as one-sided	None
Boiling points, both	Selection-aware resampling added (Section 6.4 of the main paper)	Exploratory; no verdict depends on it

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